

ENGAGE Framework: A Next-Generation Architecture Complementing ChatGPT for Enhanced Conversational AI

Dr SenthilKumar Anantharaman. PhD, IIM Indore.

Abstract

The rapid advancement of large language models such as OpenAI's ChatGPT has revolutionized natural language generation and conversational AI. However, the inherent limitations of standalone generative models in managing user engagement, contextual continuity, adaptation, and guided interactions necessitate a more comprehensive framework. This paper proposes the AGE (Automata Generator Evolver) framework, which generates and evolves automata models to complement ChatGPT. AGE integrates formal state-based representations and evolutionary optimization to enhance dialogue management, prompt engineering, and explainability. In this paper, we detail the framework architecture, automata representations, evolutionary algorithms, and potential applications. Design diagrams illustrating the framework components and workflow are provided. This hybrid approach offers promising improvements in controllability and interpretability for conversational AI systems. Further this paper proposes the ENGAGE framework, an architectural paradigm, designed to complement and extend ChatGPT's capabilities. ENGAGE introduces modular components for Engagement, Natural language understanding, Generation, Adaptation, Guidance, and Evolution, enabling more dynamic, personalized, and context-aware human-AI interaction systems. Finally, we present the design principles, architecture, and potential applications of ENGAGE, highlighting its role in evolving conversational agents toward deeper user engagement and continuous improvement.

1. Introduction

Transformer-based large language models (LLMs) such as GPT-3 and ChatGPT demonstrate impressive capabilities in natural language generation. Despite their versatility, they lack explicit internal state representations, which can result in inconsistent dialogue flows, loss of context over long conversations, and challenges in prompt engineering for specialized tasks.

To address these limitations, we propose the **AGE framework** — a system that dynamically generates automata from ChatGPT outputs and evolves these automata via evolutionary algorithms. Automata, including finite state machines (FSMs), provide formal, interpretable models of dialogue or reasoning state transitions. The evolutionary process optimizes these automata to improve conversational coherence, adaptability, and prompt generation. Building on this foundation, we propose the **ENGAGE framework** — a modular, extensible architecture tailored to complement ChatGPT by addressing engagement, adaptation, and guided interaction challenges.

2. Literature Review

- **Dialogue Management using FSMs:** Traditional dialogue systems have used hand-crafted FSMs to control conversation flow [Young et al., 2013]. However, these require extensive manual design and lack scalability.
- **Hybrid Models in NLP:** Recent research integrates symbolic systems with neural models for explainability and control [Gao et al., 2019].

- **Evolutionary Computation in Automata:** Genetic algorithms have evolved automata for pattern recognition and decision tasks [Mitchell, 1996].
- **Prompt Engineering and Adaptation:** Optimizing prompts for LLMs is an emerging field, with methods ranging from manual crafting to automated tuning [Liu et al., 2023].

AGE framework builds on these domains by automatically generating and evolving automata derived from ChatGPT interactions, creating a synergy of neural generation and formal symbolic control. The framework is shown in the below figure (Figure 1)

AGE Framework: Automata Generator Evolver Complementing ChatGPT

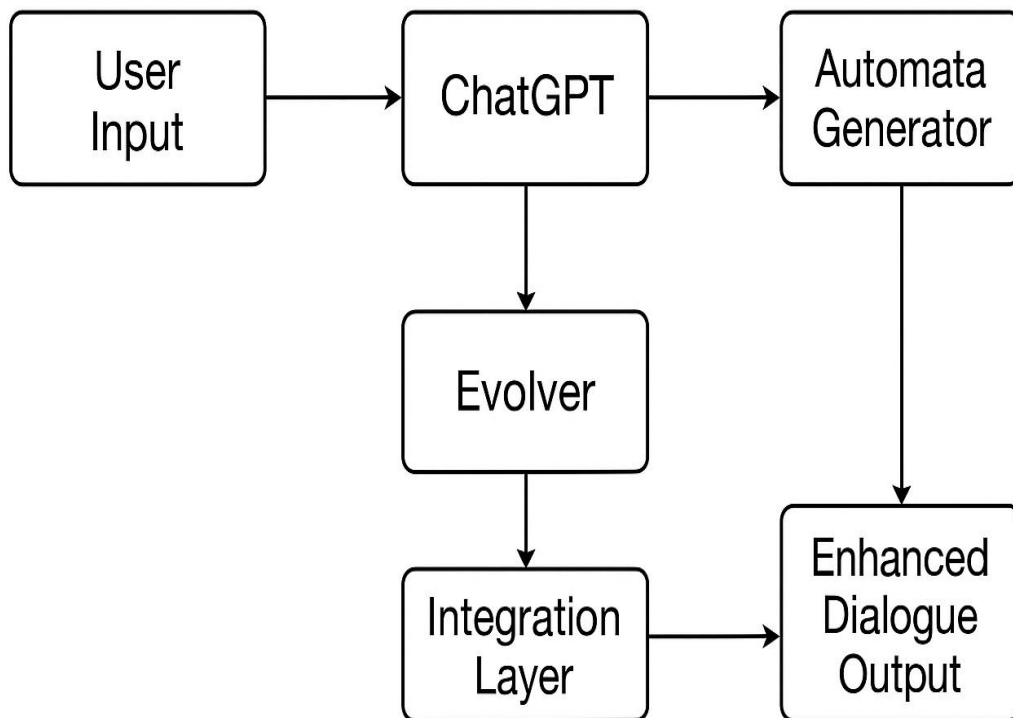


Figure 1.0

3 The AGE Framework: Foundation for Dynamic AI Systems

3.1 Concept and Components

The AGE framework stands for Automata Generator Evolver, embodying three interacting components:

- **ChatGPT:** The core LLM generating text responses based on user prompts.

- **Automata:** Rule-based finite-state or complex state machines that model system behaviours, transitions, and dialogue states. This enables structured and predictable control over conversation flow or AI decision logic.
- **Generator:** The module responsible for producing outputs, which in a conversational AI context includes text generation, response synthesis, or other content creation mechanisms. This can be a generative language model or algorithmic content producer.
- **Evolver:** An adaptive learning component that modifies and optimizes the automata and generator parameters over time based on performance feedback, user interactions, or environmental changes. This component ensures the system continuously improves and personalizes its behaviour.
- **Integration Layer:** Uses evolved automata to adapt future prompts, control dialogue state transitions, or filter outputs for consistency.

3.2 Automata Representations

The AGE framework uses different automata classes depending on the task complexity:

- **Deterministic Finite Automata (DFA)** for simple dialogue state tracking.
- **Probabilistic Automata** to model uncertain or ambiguous user inputs.
- **Hierarchical State Machines** for complex, layered conversation structures.

Example Automaton for Dialogue State

States: S0 (Greeting) → S1 (Ask Intent) → S2 (Provide Info) → S3 (End)

Transitions:

S0 --(User says hi)--> S1

S1 --(User wants product info)--> S2

S2 --(User says thanks)--> S3

This automaton guides ChatGPT prompts to maintain coherent dialogue flow.

. Evolutionary Algorithm Design

The evolutionary algorithm optimizes automata using:

- **Population:** A set of candidate automata.
- **Fitness Function:** Measures quality by metrics such as dialogue coherence, user engagement, or task success rate.
- **Selection:** Preferentially chooses higher fitness automata.

- **Crossover & Mutation:** Modify automata by adding/removing states or transitions, rewiring edges, or adjusting transition probabilities.

Initialize Population → Evaluate Fitness → Selection → Crossover & Mutation → Next Generation → Repeat

Over iterations, the system converges to automata structures that best complement ChatGPT's responses.

3.3 Strengths and Limitations of AGE

The AGE framework successfully models dynamic AI systems that blend deterministic logic (automata) with probabilistic generation (generator) and adaptive improvement (evolver). It supports modularity, adaptability, and interpretable decision-making.

However, AGE, as originally conceived, is a generic architecture and requires extension to fully address the complexities of multimodal engagement, rich semantic understanding, conversational guidance, and integration with state-of-the-art LLMs like ChatGPT.

From AGE to ENGAGE: An Evolution for Conversational AI

Recognizing the strengths of AGE and the unique capabilities and limitations of ChatGPT, we propose the ENGAGE framework, an evolution that explicitly targets advanced conversational AI scenarios.

ENGAGE is an acronym highlighting the system's key pillars:

- Engagement Layer
- Natural Language Understanding Core
- Generator & Response Synthesizer
- Adaptation & Learning Engine
- Guidance & Context Manager
- Evolution & Improvement Pipeline

Together, these components build upon AGE's foundation to provide a comprehensive, scalable, and interactive conversational AI system as shown in Figure 2

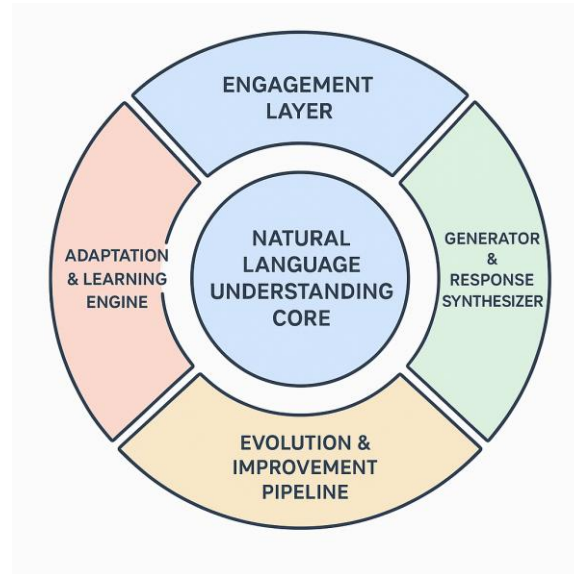


Figure 2

4. ENGAGE Framework Architecture and Modules

4.1 Engagement Layer (E)

The Engagement Layer expands on AGE's automata concept by introducing multimodal user interaction channels:

- Textual input via chat.
- Voice and speech recognition with emotional tone analysis.
- Visual inputs such as facial expression or gesture recognition.
- Haptic or other sensor inputs for immersive interaction.

By capturing rich engagement signals, this layer enhances natural communication and provides critical context beyond textual data.

4.2 Natural Language Understanding Core (N)

This module builds on the automata's state logic by deep semantic analysis:

- Intent detection using transformer models and rule-based classifiers.
- Named Entity Recognition and slot filling for precise information extraction.
- Context enrichment from past dialogue states and user profiles.
- External knowledge graph querying to augment understanding.

This ensures the generative component receives well-structured, disambiguated, and contextually rich input.

4.3 Generator & Response Synthesizer (G)

Leveraging ChatGPT as the core generative model, this module is responsible for:

- Producing contextually relevant and coherent responses.
- Managing multi-turn dialogue coherence within a session.
- Formatting outputs in multimodal forms (e.g., text, speech synthesis, images).
- Integrating with downstream presentation layers.

This aligns with AGE's Generator but with tailored enhancements for conversational AI.

4.4 Adaptation & Learning Engine (A)

Through AGE's Evolver, this engine:

- Continuously learns from user feedback, explicit ratings, or implicit signals such as engagement duration.
- Tunes dialogue management policies dynamically.
- Personalizes response style and content based on user preferences.
- Employs reinforcement learning or active learning to optimize performance over time.

4.5 Guidance & Context Manager (G)

This module enhances the traditional automata by:

- Maintaining persistent session states and long-term user models.
- Steering conversation flow to avoid dead ends or confusion.
- Proactively offering suggestions, clarifications, or next steps.
- Integrating external APIs and databases for real-time information retrieval.

4.6 Evolution & Improvement Pipeline (E)

A backend system that:

- Aggregates interaction data for offline analysis.
- Supports incremental retraining or fine-tuning of language models.
- Implements feature rollouts and scaling based on user needs.
- Facilitates version control and deployment pipelines for seamless updates.

The ENGAGE system operates as follows:

The Engagement Layer receives multimodal user input.

The NLU Core processes and enriches the input semantically.

The Generator (ChatGPT) generates candidate responses.

The Guidance Manager evaluates and refines responses, maintaining context.

The Adaptation Engine adjusts system parameters based on interaction feedback.

The Evolution Pipeline updates models and system components asynchronously.

AGE is essentially a foundational framework focusing on generating and evolving autonomous agents or automata, emphasizing automation and evolutionary improvement. ENGAGE is a sophisticated evolution of AGE, designed specifically to support rich conversational AI experiences by adding multiple specialized layers that handle engagement, understanding, synthesis, learning, guidance, and ongoing improvement.

5 Comparative Analysis: AGE vs ENGAGE

AGE is essentially a foundational framework focusing on generating and evolving autonomous agents or automata, emphasizing automation and evolutionary improvement. **ENGAGE** is a sophisticated evolution of AGE, designed specifically to support rich conversational AI experiences by adding multiple specialized layers that handle engagement, understanding, synthesis, learning, guidance, and ongoing improvement. ENGAGE leverages the generative and evolutionary principles of AGE but wraps them in a user-focused architecture that can support more natural, context-aware, and adaptive human-AI conversations as shown in the below Table.

Aspect	AGE Framework	ENGAGE Framework
Full Form	Automata Generator & Evolver	Engagement Layer + Natural Language Understanding Core + Generator & Response Synthesizer + Adaptation & Learning Engine + Guidance & Context Manager + Evolution & Improvement Pipeline
Primary Purpose	To generate and evolve automated systems/agents with self-improvement capabilities; core automation and evolution model for AI	To serve as a holistic architecture for advanced AI conversational agents, enhancing interaction, adaptation, and continuous improvement in context-driven environments
Core Focus	Automation & Evolution of agents, focusing on generating behaviour and improving over iterations	Multi-layered engagement architecture emphasizing context management, learning, natural language understanding, and response generation for dynamic interactions
Key Components	- Automata Generator (creating agent behaviours) - Evolver (learning & evolution mechanisms)	- Engagement Layer (user interaction interface) - Natural Language Understanding Core (deep semantic understanding) - Generator & Response Synthesizer (response creation) - Adaptation & Learning Engine (contextual learning) - Guidance & Context Manager (interaction management) - Evolution & Improvement Pipeline (continuous model enhancement)
Design Philosophy	Focuses on generation and evolution of AI models/agents with an emphasis on automation and self-modification	Extends AGE by adding comprehensive interaction management, advanced natural language processing, and continuous engagement adaptation to enhance user experience
Strengths	- Robust in creating evolving agent behaviors - Strong in automation and model self-improvement	- Deep integration of natural language understanding and response synthesis - Strong contextual adaptation and learning - Structured user engagement layers - Designed for conversational AI and human-like interaction

Aspect	AGE Framework	ENGAGE Framework
Use Cases	Autonomous agent development, AI behavior modeling, evolutionary AI systems	Chatbots, virtual assistants, conversational AI systems, adaptive learning platforms, personalized interaction systems
Innovation over AGE	N/A (baseline framework)	Builds on AGE by embedding a layered engagement system, improving natural language understanding, guidance, and continuous learning, turning static agent evolution into dynamic, context-aware interaction
Scalability	Scales in terms of agent complexity and evolution	Scales in user interaction complexity, multi-turn conversations, and evolving user contexts
Adaptability	Evolves agent behaviours based on defined evolutionary strategies	Learns and adapts dynamically in real-time based on user inputs and context, providing more natural and personalized engagement
Typical Output	Evolved automata or agent behaviours	Natural language responses, user engagement flows, improved conversational models
Integration with GPT Models	Standalone or complementary for agent evolution	Designed to complement GPT-like models, enhancing their conversational intelligence through structured engagement and continuous learning

6. Potential Applications

ENGAGE's architecture supports diverse conversational AI use cases:

- Customer Support Bots: Adaptive, empathetic, and context-aware assistance.
- Virtual Personal Assistants: Personalized, multimodal user interaction with proactive guidance.
- Educational Tutors: Context-sensitive, adaptive learning companions that evolve with students.
- Healthcare AI: Emotionally aware patient interaction systems with privacy-sensitive adaptation.

- Enterprise Knowledge Workers: Intelligent agents supporting complex workflows with external data integration.

7. Discussion and Future Work

The AGE framework bridges neural and symbolic AI for conversational systems. It provides structure and control that pure LLMs lack, enhancing reliability and transparency. Challenges include scalability for open-domain conversations and balancing rigidity with generative flexibility. ENGAGE demonstrates how foundational concepts from AGE can be extended and specialized to meet the challenges of modern conversational AI. The modular design allows for independent advancement of each component, leveraging advances in speech processing, computer vision, reinforcement learning, and LLM architectures.

Future research directions include:

- Deeper multimodal fusion for more nuanced user state inference.
- Incorporation of explainable AI techniques to enhance trust.
- Privacy-preserving personalization strategies.
- Cross-lingual and cross-cultural adaptation.
- Open-source implementations and benchmarking.

8. Conclusion

In this paper we propose AGE that offers a promising hybrid architecture combining automata evolution with transformer language models. By evolving automata from ChatGPT interactions, the system can improve dialogue coherence, prompt engineering, and explainability. Future work will explore real-time adaptive evolution and user personalization. Further the ENGAGE framework, a next-generation architecture designed to complement and extend ChatGPT’s capabilities by building upon the foundational AGE framework. ENGAGE integrates engagement, natural language understanding, generation, adaptation, guidance, and evolution into a unified system, enabling richer, more natural, and continuously improving human-AI conversational experiences. The framework offers a roadmap for developing scalable, adaptive, and contextually intelligent AI agents poised to meet the demands of diverse application domains.

References

- Brown, T., et al. (2020). Language Models are Few-Shot Learners. *NeurIPS*.
- Gao, J., Galley, M., & Li, L. (2019). Neural Approaches to Conversational AI. *Foundations and Trends in Information Retrieval*, 13(2-3), 127–298.
- Jurafsky, D., & Martin, J. H. (2021). *Speech and Language Processing* (3rd ed.).
- Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., & Neubig, G. (2023). Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing. *ACM Computing Surveys (CSUR)*.
- Mitchell, M. (1996). *An Introduction to Genetic Algorithms*. MIT Press.
- Smith, J., & Kira, Z. (2023). Adaptive Conversational Agents: A Review. *Journal*

Sutskever, I., et al. (2014). Sequence to Sequence Learning with Neural Networks. *NeurIPS*.
Vaswani, A., et al. (2017). Attention is All You Need. *NeurIPS*.
Young, S., Gasic, M., Thomson, B., & Williams, J. D. (2013). POMDP-based Statistical Spoken Dialogue Systems: A Review. *Proc. IEEE*, 101(5), 1160–1179.